Exercise 1 - assignment

1. $WP[x := x+5,\ x > 10]$
2. $WP[y := x-2,\ y == 0]$
3. $WP[z := y * 3,\ z \geq x]$
4. $WP[x := -x,\ x \geq 0]$
5. $WP[x := x / 2,\ x < 4]$
6. $WP[a := b + c,\ a \neq 0]$
7. $WP[x := x + 1; x := x * 2,\ x \geq 10]$
8. $WP[y := x – 1; z := y - 1,\ z == 0]$
9. $WP[a := b * 2; b := a + 3,\ b < 10]$
10. $WP[x := y; y := z; z := x,\ y == x]$

1. WP[x := x+5, x > 10]

$x+5>10$

$\boxed{x>5}$

2. WP[y := x-2, y == 0]

$x-2==0$

$\boxed{x==2}$

3. WP[z := y * 3, z ≥ x]

$\boxed{y*3 \geq x}$

4. WP[x := -x, x ≥ 0]

$-x \geq 0$

$\boxed{x \leq 0}$

5. WP[x := x / 2, x < 4]

$\frac{x}{2} < 4$

$\boxed{x<8}$

6. WP[a := b + c, a ≠ 0]

$\boxed{b+c \neq 0}$

7. WP[x := x + 1; x := x * 2, x ≥ 10]

$x*2 \geq 10$

$x \geq 5$

$x+1 \geq 5$

$\boxed{x \geq 4}$

8. WP[y := x – 1; z := y - 1, z == 0]

$y-1 == 0$

$y == 1$

$x-1 == 1$

$\boxed{x == 2}$

9. WP[a := b * 2; b := a + 3, b < 10]

$a+3 < 10$

$a < 7$

$7 = b*2$

$b = 7/2$

10. WP[x := y; y := z; z := x, y == x]

$z == y$

$z == z$

true

$x == x$

true

$\boxed{True}$

Exercise 1 – assignment /2

11. $WP[u := v + 2;\ v := u - 3,\ v \geq 0]$
12. $WP[x := x - 1;\ y := x + y,\ y > 0]$
13. $WP[\text{if }\ x >= y\ z := x - y \text{ else } z := y - x,\ z \geq 0]$
14. $WP[\text{if } x \% 2 == 0\ x := x / 2 \text{ else } x := x - 1,\ x \geq 3]$
15. $WP[\text{if } x > 10\ y := x \text{ else } y := 10,\ y > x]$
16. $WP[\text{if } x < 0\ x := -x \text{ else } x := x + 1,\ x \geq 0]$
17. $WP[\text{if } a == b\ c := 0 \text{ else } c := 1,\ c == 0]$
18. $WP[\text{if } x > y\ x := x - 1 \text{ else } x := y + 1,\ x > y]$
19. $WP[x{:=}x{+}1;\ \text{if } x > 0\ y{:=}x \text{ else } y{:=}{-x},\ y \geq 1]$
20. $WP[\text{if } x < 10\ x := x + 3 \text{ else } x := x - 3,\ x < 20]$

11. WP[u := v + 2; v := u – 3, v ≥ 0]

$u - 3 \geq 0$ $v + 2 \geq 3$

$u \geq 3$ $\boxed{v \geq 1}$

$\leftarrow$

12. WP[x := x - 1; y := x + y, y > 0]

$x + y > 0$ $x - 1 + y > 0$

$\boxed{x + y > 1}$

13. WP[if x >= y z := x – y else z := y – x, z ≥ 0]

$x - y \geq 0$ $y - x \geq 0$

$x \geq y$ $y \geq x$

$(x \geq y \rightarrow x \geq y) \wedge (x < y \rightarrow x \leq y)$

Statment is true Statment is true

14. WP[if x % 2 == 0, x := x / 2 else x := x – 1, x ≥ 3]

True

$\frac{x}{2} = \geq 3$

$x \geq 6$

False

$x-1 \geq 3$

$x \geq 4$

$(x \% 2 == 0 \rightarrow x \geq 6) \wedge (x \% 2 \neq 0 \rightarrow x \geq 4)$

15. WP[if x > 10, y := x else y := 10, y > x]

True

$x > x$

False.

$10 > x$

$x < 10$

$(x > 10 \rightarrow x > x) \wedge (x \leq 10 \rightarrow x < 10)$

false

it can not be $x > x$

we need to switch branch condition to $x \leq 10$

so from false branch ;

$x \leq 10 \rightarrow x < 10$

16. WP[if x < 0 (x := -x else x := x + 1, (x ≥ 0)]

True

$-x \geq 0$

$x \leq 0$

False

$x+1 \geq 0$

$x \geq -1$

$(x < 0 \rightarrow x \leq 0) \wedge (x \geq 0 \rightarrow x \geq -1)$

Statement is always true

Statment is always true.

17. WP[if a == b (c := 0 else c := 1), c == 0]

True

$0 = 0$

False

$1 = 0$

$(a == b \rightarrow 0 = 0) \wedge (a \neq b \rightarrow 1 = 0)$

Statment is true

Statment is false

$(a == b \rightarrow 0 = 0)$

18. WP[if x > y (x := x – 1 else x := y + 1), x > y]

True

$x-1 > y$

$x-y > 1$

$x > 1+y$

False

$y+1 > y$

$1 > y-y$

$1 > 0$

$(x > y \rightarrow x > 1+y) \wedge (x \leq y \rightarrow 0 < 1)$

Statment is ~~true~~.

Statement is false

19. WP[x:=x+1; // if x > 0 (y:=x else y:=-x,) y ≥ 1]

True

$x \geq 1$

False

$-x \geq 1$

$x \leq -1$

$(x > 0 \rightarrow x \geq 1) \wedge (x \leq 0 \rightarrow x \leq -1)$

II

$x+1 \geq 1$

$\underline{x \geq 0}$

$x+1 \leq -1$

$\underline{x \leq -2}$

$x + 1 > 0$

$\underline{x > -1}$

$x + 1 \leq 0$

$\underline{x \leq -1}$

19^{th} ans:

$(x > -1 \rightarrow x \geq 0) \wedge (x \leq -1 \rightarrow x \leq -2)$

20. WP[if x < 10 (x := x + 3 else x := x – 3) x < 20]

True

$x + 3 < 20$

$x < 17$

False

$x - 3 < 20$

$x < 23$

$(x < 10 \rightarrow x < 17) \wedge (x \geq 10 \rightarrow x < 23)$